Neel Patel

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Education

Carleton University – B.Eng, High Distinction, Computer Systems Engineering, Co-op Sept 2021 – Apr 2026
- Recipient of Entrance Scholarship, Dean's Honour List

Experience

AI/ML Research Assistant | Carleton Immersive Media Studio – Ottawa, ON May 2026 – Present
- Technical team lead of an AI-driven web-scraping system designed to handle 50+ unique websites
- Implemented a robust LLM-based schema mapper using vLLM, aggresively optimizing to run 54 parallel instances of a 4B model within 20GB of VRAM
- Currently enabling digital twin technologies with computer vision pipelines

Software Developer Student Co-op | Solace – Ottawa, ON Winter 2024, Fall 2024, Summer 2025
- Streamlined SVN to Git migration, cutting a multi-week process down to a single-day task
- Architected, implemented, and tested features for Solace C and Python APIs, optimizing bandwidth usage and enabling message compression
- Led the migration and open-source release of the Wireshark SMF Plugin, improving community initiatives
- Automated CI/CD pipelines and releases using GitHub Actions and Jenkins, enabling automated developer QA and plugin deployment

Full-Stack Developer Co-op | LearnExperts – Ottawa, ON Summer 2024
- Automated testing and report generation with GitHub Actions, increasing bug visibility and reducing regressions
- Enhanced bug tracking by implementing AWS RUM with custom metadata for error logs
- Developed end-to-end Cypress tests for key interactive workflows, reducing manual test requirements by 40%

Jr. Test Analyst Co-op | Immigration, Refugees, and Citizenship Canada – Ottawa, ON Summer 2023
- Ensured reliability of government application used by 10 000+ employees through verification of manual tests
- Collaborated with IT and development teams to formalize and document test procedures, improving traceability

Jr. Software Developer, High School Co-op | NorthStar Utilities – Ottawa, ON Winter 2020
- Standardized customer .properties files by developing a Java utility that parses and cleans data
- Leveraged Git, Docker, and agile methodology, gaining early experience in developer practices

Projects

- **Wallpaper Theme Converter:** JavaScript website that changes the colour scheme of wallpapers, with 170+ stars on GitHub
- **AI Capstone Project:** Architected a multi-agent RAG system to analyze City of Ottawa's asset management documents
- **AI Raspberry Pi Bamboo Plant:** Engineered an IoT monitoring system with AI voice capabilities to monitor water levels and generate personalized voice alerts
- **Ross Video FPGA Challenge:** Designed a SystemVerilog and Python system to simulate GIF rendering via a custom ROM for the FPGA, winning team
- **FPGA Colour Filter:** Used Zybo Z7-10 + Verilog to develop a colour filter LUT system with HDMI output

Skills

Languages: Python, Java, C, C++, ARM Assembly, Verilog, BASH, HTML, CSS, JavaScript
Project Management: Git, SVN, Jira, Confluence
Hardware/Embedded: Vivado, Zybo FPGA, Raspberry Pi, Arduino
Infrastructure: GitHub Actions, Jenkins, Docker, Azure
Other Tools: Claude Code, Agentic AI, Wireshark, GDB, Web Scraping, Computer Vision